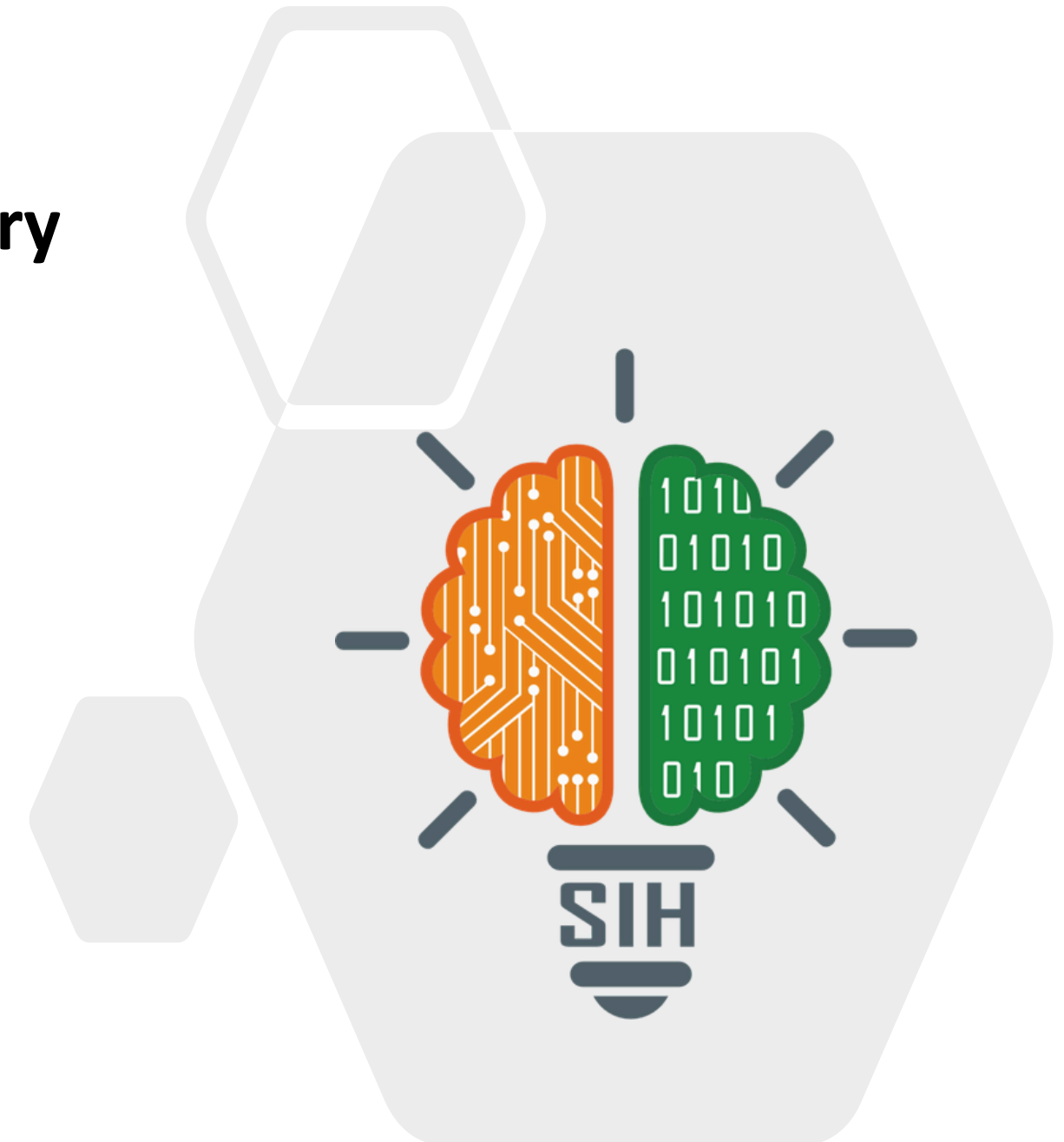


ECDAT

Enterprise Cryptographic Discovery & Analysis Tool

- Problem Statement ID – SIH26164
- Problem Statement Title – Enterprise Cryptographic Discovery & Analysis Tool (ECDAT)
- Theme – Blockchain & Cybersecurity
- PS Category – Software
- Team ID – [ADD TEAM ID]
- Team Name – Astarr



PROPOSED SOLUTION

India's critical systems run on RSA & ECDSA. Quantum computers will break them by 2031. ECDAT finds every vulnerability — before it's too late.

THE PROBLEM



SHOR'S THREAT:

Breaks RSA-2048 & ECC powering defense, banking, and PKI.



“Harvest Now, Decrypt Later”

Adversaries are stealing encrypted data TODAY to decrypt it when quantum computers arrive.

OUR SOLUTION



Multi-Vector Discovery

Scans source code (Python, Java), X.509 certificates, server configs, and live TLS 443 endpoints.



Mathematical Threat Scoring

QARS (0–100) + Mosca's Theorem tells you exactly how urgent each threat is.



1-Click Sovereign Fix

Generates NIST-certified ML-KEM / ML-DSA replacement code and a full CBOM audit report.

WHO IS IT FOR?



NTRO Analyst

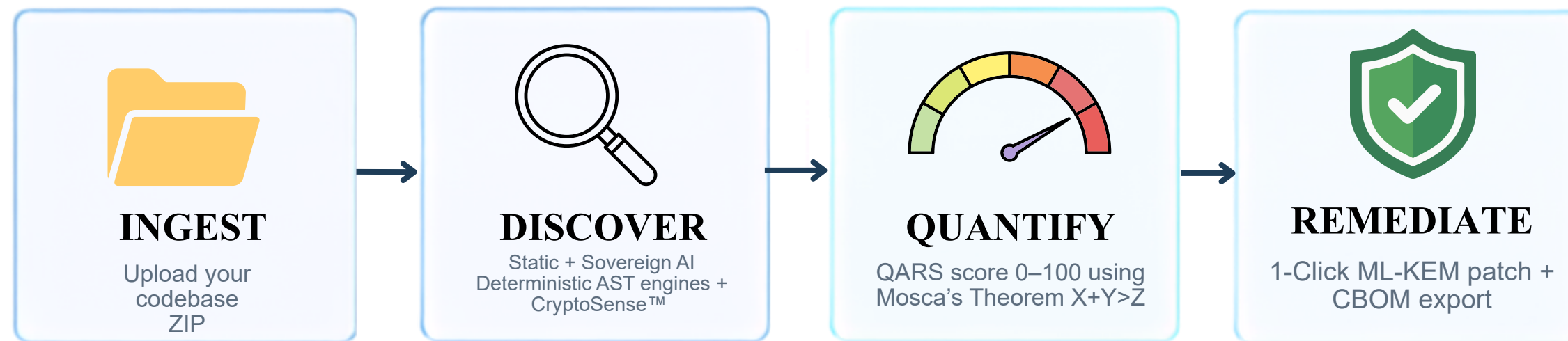


CISO /
Security Lead



DevSecOps
Engineer

HOW IT WORKS — END-TO-END PIPELINE



Migration takes 4–7 years. Under Mosca's Theorem ($X+Y>Z$), India's critical data is already inside the HNDL breach window.

INNOVATION & UNIQUENESS



CryptoSense™ AI

Local CodeBERT finds hidden wrappers.
100% air-gapped — zero cloud leaks



1-Click Patch

Automated Git diffs: RSA/ECC → ML-KEM/ML-DSA. Fully compliant with NIST FIPS 203/204.

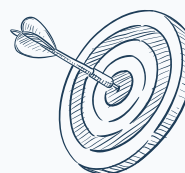


Mosca Urgency Engine

Real-time $X + Y > Z$ timeline mapping active HNDL breach windows



**Minutes Not
Weeks**



**AI + AST
Hybrid**



**NIST FIPS
203/204/205**



**Built For
India**

TIER 01

Multi-Vector Ingestion

TIER 02

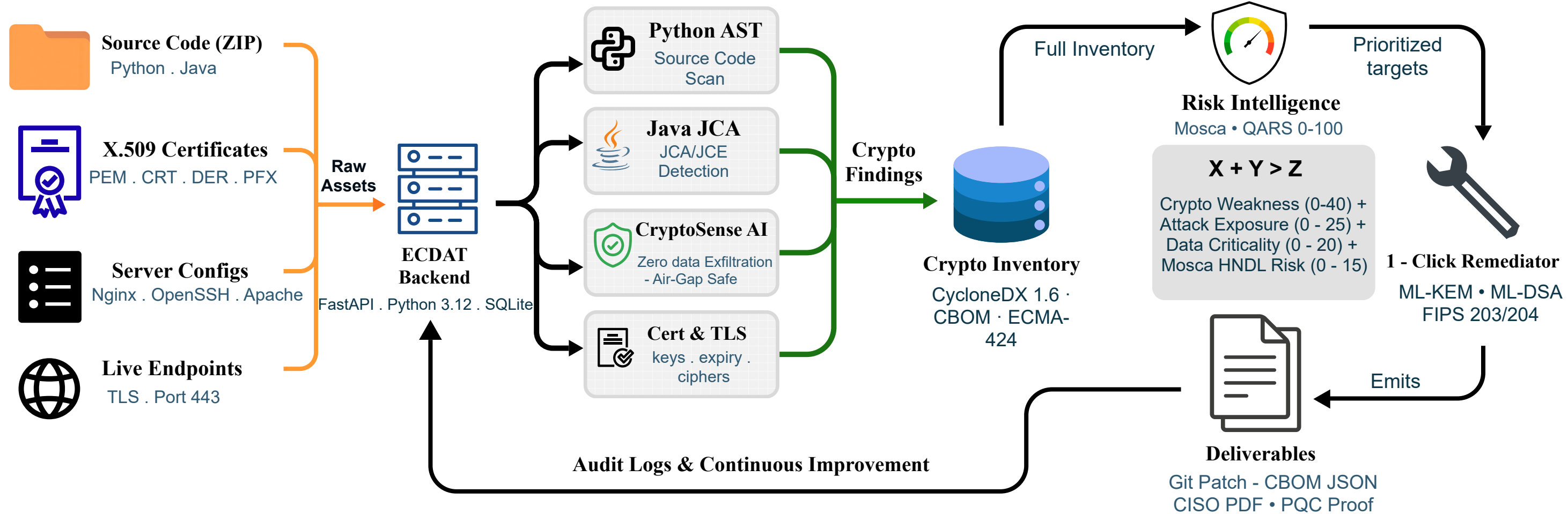
Static & AI Discovery Engines

TIER 03

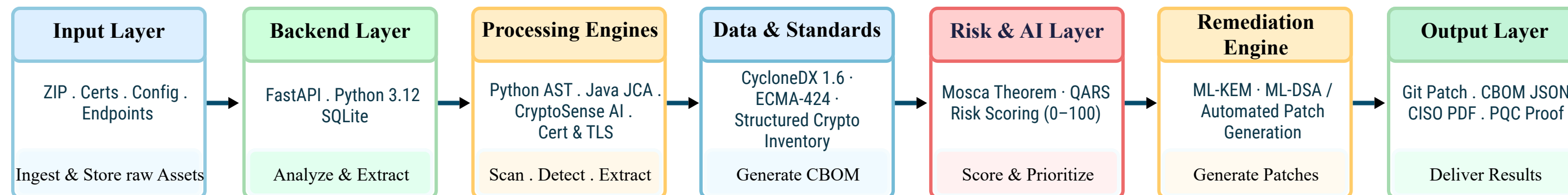
Inventory & Risk Intelligence

TIER 04

Remediation & Deliverables

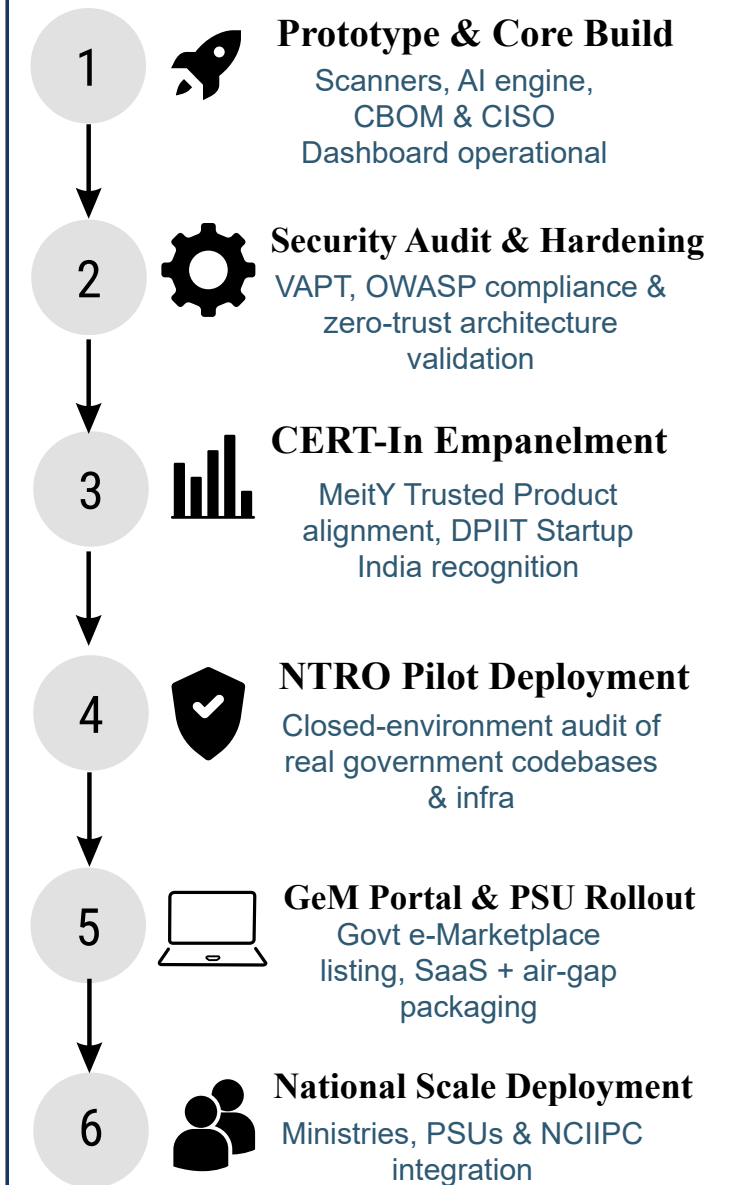


TECH STACK WORKFLOW



IMPLEMENTATION PLAN

SIH Prototype → Sovereign National Deployment



Outcome

 **Prototype - [Click Here](#)**
Repo. - [Click Here](#)
Report - [Click Here](#)

FEASIBILITY

**Technical Feasibility:**

Lightweight Python/FastAPI core; native AST static parsing runs 100% locally with zero external runtime dependencies.

**Operational & Air-Gap Ready**

Executes 100% on-premise within classified defense networks; zero outbound telemetry or cloud data exfiltration.

**Standard Compliance**

Strict compliance with NIST FIPS 203/204/205, CycloneDX 1.6 (ECMA-424 CBOM), and CERT-In quantum guidelines

**Implementation Readiness**

Production-grade prototype; 71/71 automated tests passing (100%), verified live on Vercel & Render



BUILD | SECURE | DEPLOY

VIABILITY

**National Mission Alignment:**

Directly empowers India's National Quantum Mission; eliminates strategic dependency on foreign proprietary tools (IBM, SandboxAQ).

**Economic Viability**

Open-core architecture with local AI eliminates millions in recurring enterprise per-seat and SaaS token licensing.

**Rapid 90-Day Deployment:**

Production-ready containerized pipeline ready for immediate pilot rollouts across critical defense and banking repositories

**Enterprise Viability**

Headless CLI + API seamlessly integrates into GitHub Actions, GitLab CI, and Jenkins as an automated PR security gate



EFFICIENT | IMPACTFUL | NATION-READY

CHALLENGES & RISKS

**Obfuscated Cryptography:**

Static AST and regex tools miss dynamic runtime imports and nested custom wrapper classes.

**Monolith Scalability:**

Scanning 1M+ LOC enterprise repositories risks RAM exhaustion and system timeouts

**System Breakage During PQC:**

Hard-swapping legacy RSA/ECC directly to PQC breaks backward compatibility with legacy clients

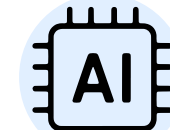
**Alert Fatigue & Noise**

Vague, unprioritized scanner reports flood security teams with hundreds of false-positive alerts



ANTICIPATE | MITIGATE | STAY AHEAD

MITIGATION STRATEGIES

**CryptoSense Sovereign AI:**

Fine-tuned on-premise CodeBERT inspects suspicious function semantics with zero cloud data exfiltration.

**Chunked Stream Buffers:**

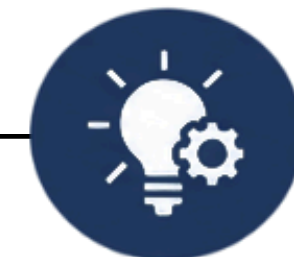
1MB chunked file streaming and async AST parsing ensure a flat <128MB RAM footprint even on massive repos

**Hybrid Migration Guidance:**

Recommends transitional hybrid pairs (X25519 + ML-KEM-768) to ensure zero client downtime during migration.

**QARS Composite Scoring:**

0–100 deterministic scoring with Mosca's Theorem ($X+Y>Z$) filters out noise, highlighting only urgent HNDL assets



SOLVE | OPTIMIZE | ENABLE SCALE

1. STRATEGIC NATIONAL IMPACT



HNDL THREAT IMMUNITY

- Pre-empts nation-state adversaries harvesting encrypted defense data today for post-quantum decryption.



CRYPTOGRAPHIC SUPPLY CHAIN

- First verifiable national standard for India's sovereign software repositories.



₹6,003 CR NQM ALIGNMENT

- Direct operational alignment with India's National Quantum Mission 2023 – 2031 sovereign defense mandate.

2. WHO BENEFITS AND HOW



DEFENSE & NTRO

- 100% air-gapped execution.
- Auto-inventories classified military crypto assets.



BANKING & NPCI

- 14.4B+ monthly UPI shielded.
- One-click CycloneDX 1.6 CBOM audit export.



CRITICAL INFRASTRUCTURE

- Secures power grid SCADA control systems.
- Protects satellite & space telemetry encryption.



DEVSECOPS & ENGINEERS

- < 10s automated discovery.
- 1-Click NIST PQC Git patches.

3. SOVEREIGN ECONOMIC VALUE



FOREX RETENTION (₹15 – 20 CR)

- Eliminates IBM/Sonar/SandboxAQ foreign USD licensing dependency.
- Eliminates recurring vendor licensing costs.



ZERO-COST NATIONAL ROLLOUT

- Free open-core for NTRO/PMO/Defense
- Paid enterprise license for regulated banks & PSUs.
- 100% sovereign — built in India, for India.



WE FIX, THEY ONLY FLAG

- 1-Click NIST PQC Git Diffs (IBM/Sonar give no patches).
- 100% Air-Gapped CryptoSense™ AI (zero cloud exfiltration)

APPLIED & CRYPTOGRAPHIC FIELD RESEARCH

- **Production Codebase Benchmarking:**
Audited 5 production-grade open-source repositories (liboqs-python, paramiko, pyjwt, oauthlib, tink-crypto) to map active Harvest Now, Decrypt Later (HNDL) cryptographic debt across legacy RSA-2048 and ECC implementations.
- **Automated Testbench & Schema Validation:**
Executed 71/71 passing unit and integration tests verifying Python/Java AST scanners, X.509 certificate parsers, and CycloneDX 1.6 CBOM schema compliance.
<https://github.com/sps-exe/ecdat>

ACADEMIC & GOVERNMENT SOURCES

- **NIST Post-Quantum Cryptography Final Standards (August 13, 2024):**
Official Federal Information Processing Standards — FIPS 203 (ML-KEM), FIPS 204 (ML-DSA), and FIPS 205 (SLH-DSA) specifying algorithms to replace quantum-vulnerable public-key cryptography. <https://csrc.nist.gov/projects/post-quantum-cryptography>
- **National Quantum Mission (NQM) — Department of Science & Technology, Govt. of India (2023–2031):**
Official government policy and operational guidelines for quantum communications, cryptography transition, and national security preparedness.
<https://dst.gov.in/national-quantum-mission-nqm>
- **Mosca's Theorem ($X + Y > Z$) Quantum Risk Framework:**
Dr. Michele Mosca (Oxford / University of Waterloo) — “Cybersecurity in an Era with Quantum Computers: Will We Be Ready?” providing the mathematical foundation for HNDL risk quantification. <https://eprint.iacr.org/2015/1075.pdf>
- **CycloneDX 1.6 Cryptographic Bill of Materials (CBOM) Standard (ECMA-424, 2024):**
Standardized machine-readable bill of materials specification defining cryptoProperties for sovereign cryptographic asset visibility.
<https://cyclonedx.org/capabilities/cbom/>
- **CERT-In & NTRO Cyber Security Guidelines:**
Adherence to NCIIPC Guidelines for Critical Information Infrastructure, CERT-In Cyber Security Directions, and National Cyber Security Policy standards.
<https://www.cert-in.org.in/>